

Year 7 • Lesson 10.04 Stem-and-leaf plots: answers (teacher copy)

10.04 Stem-and-leaf plots (booklet p. 3)

Warm-up

- 4
- 3
- 1, 2, 6, 8
- 7

WE DO examples

- Ex 1: 20, 23, 23, 27
- Ex 2: 10 people; youngest 12, oldest 46
- Ex 3: 6 | 2 5 8, 7 | 0 1 3 9, 8 | 0 4 4
- Ex 4: 4 | 3 7 8, 5 | 5 5 6 9, 6 | 1 2 (key 4 | 3 = 43)
- Ex 5: 13 | 8 8, 14 | 2 7 9, 15 | 1 3 5, 16 | 0 (key 14 | 2 = 142 cm)
- Ex 6: 2 | 4 4 8, 3 | 1 5 9, 4 | 0 (key 2 | 4 = 2.4)

Set A

- 14
- 24
- 50
- 43
- 5
- 7
- 3
- 8
- 2
- 46

Set B

- 4 | 7
- 6 | 3
- 10 | 5
- 0 | 9
- 1 2 6 8
- 0 3 3 4 9
- 2 | 2 3 6 7, 3 | 1 1 5 8, 4 | 0 4
- 4
- 31
- 44

Set C

- Ordered: 7 | 1 5 6 8 9; 8 | 0 0 2 3 4 5 8; 9 | 0 2 2 2 4 8; 10 | 4 5
- 20 players
- 71 kg
- 105 kg
- 92 kg
- 5
- 8
- $\frac{7}{20}$
- $\frac{2}{20} = 10\%$
- You can see the lowest, highest and most common values straight away

Set D

- 13 | 5 8 8, 14 | 2 2 2 6 9, 15 | 1 4, 16 | 0 (key 13 | 5 = 13.5 s)
- 11
- 13.5 s
- 14.2 s
- 3
- $\frac{3}{11}$
- 2.5 s
- 10 | 9, 11 | 2 8, 12 | 4 4 7, 13 | 1 5, 14 | 0 (key 12 | 4 = 124)
- 10 to 14: the first two digits, so every leaf is one digit
- So the reader knows what the numbers mean (e.g. 2 | 4 could be 24 or 2.4)
- It shows every actual value, and still shows the shape

Extension

- 2 | 0 1 3, 2* | 5 6 8, 3 | 1 2, 3* | 7
- 0.67
- When one stem has many leaves, so the data is bunched up

Exit ticket

- Level 1: 8
- Level 2: 4 | 0 2 6 6
- Level 3: $\frac{5}{8}$; mode 46